AI-Powered Cyber Attacks

Cyber Security Semester Executive Summary Report

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Artificial intelligence has become a dual-use force in cybersecurity. The same generative models that support threat intelligence, secure coding, anomaly detection, and incident response can also help adversaries produce convincing phishing messages, automate reconnaissance, discover vulnerabilities, generate malicious code, create deepfake identities, and adapt attacks at machine speed. Recent research therefore treats AI-powered cyber attacks not as one isolated attack category, but as an acceleration layer that increases the scale, personalization, speed, and evasiveness of existing attack techniques (Coppolino et al., 2025; Ferrag et al., 2025; Zhang et al., 2025).

This report reviews ten peer-reviewed journal articles published in 2025–2026 by IEEE, ACM, Springer, and Elsevier. The reviewed literature covers cross-industry generative-AI risks, LLM-assisted phishing, AI-supported vulnerability exploitation, UAV jamming and spoofing, deepfake-enabled cybercrime, and AI-based detection and prediction measures. A central result is that AI reduces the expertise and time required to produce credible attacks while also enabling defenders to automate analysis. The security outcome therefore depends on governance, data quality, model access controls, human validation, and the ability of defensive systems to generalize to previously unseen AI-generated content.

The review was limited to journal articles whose final publication year is 2025 or 2026. Studies were selected for direct relevance to offensive AI use, measurable attack consequences, or defensive measures against AI-amplified threats. Each paper was compared using the required dimensions: industry, attack, impact, measures, and future directions. The table on Pages 3–4 presents the comparative synthesis, while Pages 5–6 summarize the major findings, recommendations, conclusion, and APA references.

Scope statement: This report focuses on the malicious or adversarial use of AI in cyber operations. Defensive AI is discussed only where it directly addresses the reviewed AI-powered attacks.

1. Topic Overview

1.1 Meaning of AI-Powered Cyber Attacks

AI-powered cyber attacks are malicious operations in which machine learning, generative AI, large language models (LLMs), deepfake systems, or autonomous agents improve one or more stages of an attack. The AI component may generate content, select targets, discover weaknesses, produce code, imitate trusted people, evade detection, or continuously refine an attack strategy. The reviewed studies show that the most important change is not the creation of completely new attack families; rather, AI strengthens familiar techniques such as phishing, malware, spoofing, social engineering, vulnerability exploitation, and disinformation (Coppolino et al., 2025; Ferrag et al., 2025; Mirtaheri et al., 2026).

1.2 How AI Changes the Attack Lifecycle

Reconnaissance and targeting. LLMs can summarize public information and transform it into personalized lures. In a large organizational study, LLM-generated lateral-phishing emails were found to be comparable in effectiveness to professionally written messages, demonstrating that automation can preserve persuasive quality at scale (Bethany et al., 2025).

Content and payload generation. Generative models can create grammatically correct phishing emails, synthetic identities, malicious instructions, code fragments, and realistic audio or video. This reduces linguistic errors and makes impersonation more credible. Deepfakes further expand social engineering from text-only deception to multimodal identity fraud and disinformation (James et al., 2026; Sivaneswaran et al., 2026).

Exploitation and adaptation. AI-assisted vulnerability assessment can accelerate discovery, prioritization, and possible exploitation of software weaknesses. However, models may also hallucinate, mis-rank risk, leak sensitive data, or be manipulated through prompt injection and poisoned inputs. Human validation and secure model integration are therefore required throughout the vulnerability-management lifecycle (Mirtaheri et al., 2026; Zhang et al., 2025).

Physical and cyber-physical effects. In unmanned aerial vehicle environments, generative methods can support sophisticated jamming and spoofing scenarios that threaten navigation, communication, and mission continuity. This illustrates how AI-powered attacks may move beyond information theft and cause operational or safety consequences (Sarikaya & Bahtiyar, 2025).

1.3 Main Risk Characteristics

Across the literature, five characteristics repeatedly appear: *automation*, *personalization*, *lower technical barriers*, *rapid adaptation*, and *high-quality synthetic content*. These characteristics increase the probability that traditional signature-based or rule-based controls will fail. At the same time, defensive research shows that layered systems combining LLM or multimodal reasoning, conventional machine learning, diverse phishing indicators, adversarial testing, user awareness, and human oversight can improve resilience (Ankalaki et al., 2025; Kulkarni et al., 2025).

1.4 Affected Industries

The threat is cross-sector. Finance and e-commerce face fraud and credential theft; education and enterprises face lateral phishing; healthcare and public services face impersonation, privacy breaches, and service disruption; software and cloud providers face automated vulnerability exploitation; media and government face deepfake disinformation; and aviation or critical infrastructure face cyber-physical spoofing and jamming. The risk is therefore both technical and organizational, requiring coordinated controls across people, process, technology, and policy.

The comparative evidence from the ten selected papers is presented on the following two pages.

2. Comparative Analysis Table (Part I)

Study / Industry	AI-Powered Attack	Impact	Measures	Future Directions	Critical Observation
Ferrag et al. (2025) Cross-industry, IoT, software, hardware	LLM misuse, prompt injection, insecure output, data poisoning, malicious code and phishing support.	Wider attack surface, confidentiality and integrity loss, automated exploitation, unsafe model outputs.	Secure model lifecycle, input/output controls, adversarial testing, data governance, access control, human review.	Cybersecurity-specific benchmarks, stronger evaluation datasets, secure fine-tuning and real-time safeguards.	Broadest taxonomy in the sample; useful for connecting model vulnerabilities with operational cyber threats.
Coppolino et al. (2025) Network and web security	Generative models used to create evasive traffic, synthetic attack patterns, deceptive web content, and adaptive attacks.	Detection bypass, faster attack variation, reduced reliability of static signatures, increased web and network risk.	Adversarial training, continuous monitoring, provenance checks, behavior-based detection, periodic model updating.	Common test environments and realistic evaluations of offensive and defensive generative models.	Emphasizes that GenAI strengthens both attackers and defenders, creating a continuing attack–defense cycle.
Zhang et al. (2025) Cybersecurity operations across sectors	LLM-assisted vulnerability analysis, malicious-code support, phishing, and manipulation of cybersecurity workflows.	Dual-use risk, privacy leakage, hallucinated security advice, model manipulation, and unsafe automation.	Human-in-the-loop review, specialized cyber LLMs, secure training data, robust evaluation, prompt and tool controls.	Better cybersecurity benchmarks, trustworthy agents, explainability, privacy protection, and reproducible evaluation.	Reviews more than 300 works and shows that model construction and downstream deployment must be secured together.
Mirtaheri et al. (2026) Software, cloud, vulnerability management	AI-assisted reconnaissance, vulnerability discovery, prioritization, exploit support, and automated remediation misuse.	Faster exploitation, incorrect risk ranking, unsafe automated actions, and expanded exposure of software assets.	Explainable risk scoring, human validation, secure SOAR integration, audit trails, red-team testing, least privilege.	Auditable end-to-end vulnerability management, governance standards, trustworthy autonomous remediation.	Demonstrates that the same GenAI capability can improve remediation or accelerate exploitation depending on controls.
Sarikaya & Bahtiyar (2025) UAVs, aviation, critical infrastructure	GenAI-based radio jamming and spoofing scenarios targeting navigation and communication signals.	Navigation errors, loss of control, mission disruption, safety hazards, and cyber-physical damage.	Signal authentication, multi-sensor validation, anomaly detection, resilient communication, adversarial training.	Real-world datasets, adaptive countermeasures, secure autonomous coordination, and field validation.	Extends AI-powered cyber risk from digital information systems to physical safety and operational continuity.

Sources: (Coppolino et al., 2025; Ferrag et al., 2025; Mirtaheri et al., 2026; Sarikaya & Bahtiyar, 2025; Zhang et al., 2025). Measures and future directions are synthesized from each study's analysis and recommendations.

2. Comparative Analysis Table (Part II)

Study / Industry	AI-Powered Attack	Impact	Measures	Future Directions	Critical Observation
Bethany et al. (2025) Education and large enterprises	Personalized LLM-generated lateral-phishing emails that imitate trusted internal communication.	Credential compromise, internal account abuse, data exposure, and increased click risk across organizational roles.	Role-based awareness, phishing simulation, MFA, email authentication, rapid reporting, behavioral detection.	Larger cross-sector experiments, role-specific risk models, and detectors that generalize across different LLMs.	Real-world study of about 9,000 organizational users found LLM-written messages comparable to professional writing.
Ankalaki et al. (2025) Cross-industry and critical systems	Evolving cyber attacks supported by ML, deep learning, and generative techniques; focus on prediction of attack behavior.	Faster and more complex attacks, delayed response when models use incomplete, imbalanced, or outdated data.	Hybrid predictive models, updated datasets, anomaly detection, feature engineering, explainable outputs.	Real-time and explainable prediction, cross-domain datasets, improved handling of unseen and zero-day behavior.	Shows that prediction capability is a central defensive requirement when adversaries can rapidly change attack patterns.
Kulkarni et al. (2025) Finance, e-commerce, and online services	Adversarial phishing webpages generated with diverse visual and structural features to evade detection models and deceive users.	Detector bypass, credential theft, brand impersonation, and reduced trust in automated phishing protection.	Multimodal LLM analysis, adversarial testing, diverse benchmarks, and combined visual, URL, and webpage-structure indicators.	Cross-brand and cross-model evaluation, adaptive attack datasets, explainable decisions, and continuous robustness testing.	PhishOracle-generated pages evaded established detectors and deceived users, while the evaluated multimodal LLM was comparatively robust.
Sivaneswaran et al. (2026) Finance, enterprise, public services	Automated, personalized, and increasingly multimodal phishing generated or improved by LLMs.	Greater scale and credibility, financial and privacy losses, reduced effectiveness of grammar-based warning signs.	LLM-based detection, user education, content and sender verification, multimodal analysis, model comparison.	Cross-model generalization, multimodal benchmarks, real-world datasets, and evaluation of adaptive attacker behavior.	Finds that LLMs accelerate both phishing generation and detection, but detection performance may not transfer across models.
James et al. (2026) Finance, media, government, justice	Deepfake audio/video/image impersonation used for fraud, social engineering, identity theft, and disinformation.	Financial, psychological, reputational, legal, and societal harm; erosion of trust in digital evidence.	Deepfake forensics, media provenance, identity verification, zero-trust procedures, legal and policy safeguards.	Real-time multimodal detection, shared datasets, stronger case studies, policy harmonization, public awareness.	Deepfakes convert identity and trust into attack surfaces and can bypass controls that assume media authenticity.

Sources: (Ankalaki et al., 2025; Bethany et al., 2025; James et al., 2026; Kulkarni et al., 2025; Sivaneswaran et al., 2026). The table is an analytical synthesis rather than a verbatim reproduction of any one paper.

3. Key Findings

1. **AI mainly amplifies established attack methods.** The strongest evidence concerns phishing, social engineering, spoofing, malicious-code support, vulnerability exploitation, and deepfake impersonation. AI makes these attacks faster, cheaper, more personalized, and easier to scale (Bethany et al., 2025; Ferrag et al., 2025; James et al., 2026).
2. **Persuasive quality is becoming less dependent on attacker skill.** LLMs remove many spelling, grammar, and style indicators traditionally associated with phishing. The large-organization experiment by Bethany et al. (2025) shows that AI-generated phishing can perform at a level comparable to professionally written communication.
3. **The attack surface now includes the AI system itself.** Prompt injection, poisoned training or retrieval data, insecure tool use, unsafe outputs, privacy leakage, and model manipulation can turn a defensive AI system into an attack path (Ferrag et al., 2025; Zhang et al., 2025).
4. **Multimodal attacks increase trust manipulation.** Deepfake audio and video allow adversaries to imitate people, organizations, and events. This creates risks that cannot be addressed by text filtering alone and requires identity verification and media provenance controls (James et al., 2026; Sivaneswaran et al., 2026).
5. **AI-powered attacks may create physical consequences.** UAV jamming and spoofing research demonstrates that AI-supported attacks can affect navigation, communication, safety, and mission continuity, especially in autonomous and cyber-physical systems (Sarikaya & Bahtiyar, 2025).
6. **No single defensive model is sufficient.** Effective defense requires layered controls: secure model engineering, conventional security controls, behavioral analytics, user education, MFA, provenance, explainability, and human authorization for high-impact actions (Ankalaki et al., 2025; Kulkarni et al., 2025; Mirtaheri et al., 2026).

4. Recommended Measures

- **Protect identities and communication:** enforce MFA, email authentication, out-of-band verification for financial or privileged requests, and role-specific phishing exercises.
- **Secure the AI lifecycle:** control model and tool access, validate training and retrieval data, filter inputs and outputs, log model actions, and red-team deployed systems.
- **Use layered detection:** combine signature, behavioral, contextual, and AI-based detection; include text, URL, audio, image, video, and network indicators where relevant.
- **Keep humans responsible for high-impact actions:** require approval for remediation, account changes, money transfers, code deployment, or safety-critical autonomous decisions.
- **Prepare for evidence and trust attacks:** adopt media provenance, strong identity verification, incident playbooks for deepfake impersonation, and procedures for preserving forensic evidence.

5. Future Research Directions

The most urgent research need is evaluation under realistic attacker adaptation. Future datasets should include current LLM-generated phishing, multilingual content, multimodal deepfakes, and attacks produced by different model families. Defensive models must be tested for cross-model generalization instead of only on the model used during training. Research should also develop auditable AI agents, standard risk benchmarks, privacy-preserving data sharing, and safe human–AI workflows. For cyber-physical systems, laboratory findings should be validated in controlled real-world environments before autonomous defense is trusted.

Overall interpretation: AI is changing the economics and speed of cybercrime. Security programs must therefore reduce both technical exposure and misplaced human trust, while ensuring that defensive automation remains transparent, constrained, and accountable.

6. Conclusion

The reviewed 2025–2026 literature confirms that AI-powered cyber attacks represent a significant evolution in the threat landscape. Generative AI and LLMs enable attackers to produce credible phishing content, support vulnerability discovery and malicious coding, automate targeting, and imitate trusted identities through deepfakes. In cyber-physical environments, AI-supported spoofing and jamming can also threaten operational safety. These capabilities lower entry barriers and allow attacks to be adapted and repeated at a scale that manual methods cannot easily match.

However, the literature does not support the conclusion that AI automatically gives attackers an absolute advantage. The same technologies can improve detection, prediction, threat analysis, and response. The decisive factor is whether organizations deploy AI within a secure and governed architecture. Layered controls, high-quality data, human validation, identity verification, explainability, and continuous testing are repeatedly identified as necessary safeguards. Future cybersecurity practice must therefore treat AI as both a defensive capability and a protected high-risk asset. Organizations that secure only their networks but not their AI models, data pipelines, automated tools, and human decision processes will remain vulnerable to the next generation of adaptive attacks.

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